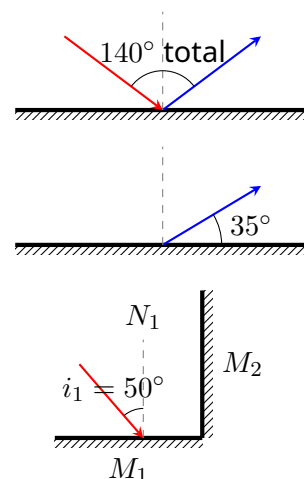


Instructions: Write all your justifications, calculations, and ray diagrams on a separate sheet of paper.

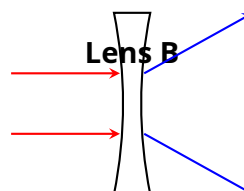
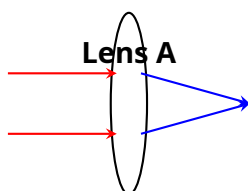
Exercise 1 : Reflection Angles & Mirror Systems

- On a plane mirror, the incident ray and the reflected ray make a combined angle of 140° together. Calculate the angle of incidence (i) and the angle of reflection (r).
- A light ray strikes a plane mirror such that the **reflected ray** makes an angle of 35° with the **surface of the mirror**. Calculate the angle of reflection (r) and the angle of incidence (i).
- Two-Mirror System:** Two plane mirrors M_1 and M_2 are placed perpendicular to each other. An incident ray strikes M_1 with an angle of incidence $i_1 = 50^\circ$. It reflects off M_1 and then strikes M_2 . Calculate the angle of reflection r_2 on mirror M_2 . Show your geometric steps clearly.



Exercise 2 : Light Beams and Lenses

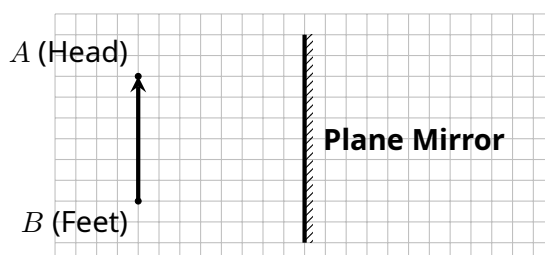
The figures below show a cylindrical (parallel) light beam falling onto two different optical systems: Lens A and Lens B. **Identify** the nature of the incident beam and the emerging beam for each case (Choose from: *Parallel, Converging, Diverging*).



Incident beam: ; Emerging beam: Incident beam: ; Emerging beam:

Exercise 3 : Image Formation by a Plane Mirror

- State whether the image formed in the following scenarios is **Real** or **Virtual**:
 - The image captured on a digital camera's internal sensor.
 - The reflection of your face seen in a bathroom mirror.
- A person is standing in front of a plane mirror. The person is represented by the object arrow AB in the grid below, where A is the head and B is the feet.
 - On your separate sheet (or by copying the grid), draw a ray diagram to find the image A' of point A . You must draw two incident rays from A (one normal and one oblique) and extend their reflections.
 - Deduce the location of the image B' for the feet, and draw the entire virtual image $A'B'$ of the person.
 - Based on your diagram, conclude three main characteristics of the image formed by the plane mirror.



Exercise 4 : The Scenic Airplane Trip

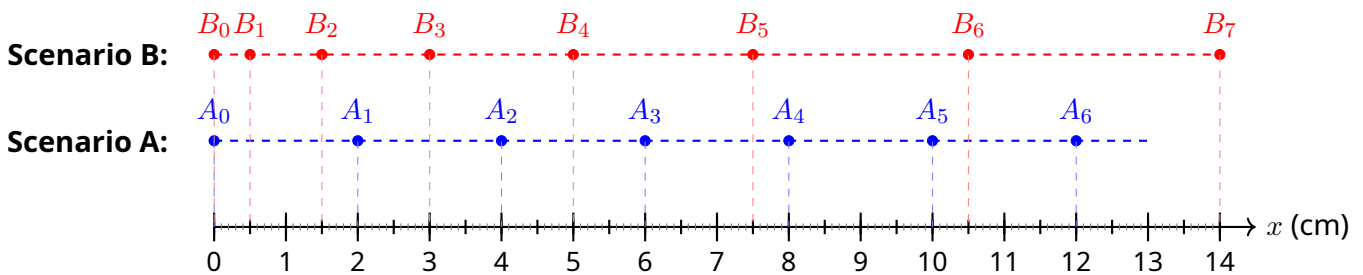
An airplane embarks on a multi-phase journey across the country:

- **Phase 1:** The plane flies at a cruising speed of $V_1 = 600$ km/h for a duration of $\Delta t_1 = 2$ hours.
- **Phase 2:** The plane slows down to allow passengers to view a scenic mountain range, covering a distance of $d_2 = 50$ km at a reduced speed of $V_2 = 200$ km/h.
- **Phase 3:** The plane resumes its journey at $V_3 = 750$ km/h to cover the final distance of $d_3 = 600$ km.

Calculate the total distance (d_{tot}), the total duration (Δt_{tot}) in hours, and deduce the average speed of the airplane for the entire trip (V_{avg}) in **km/h** and then convert it to **m/s**.

Exercise 5 : Motion Recording Analysis

The figure below shows two chronophotographic recordings (Scenario A and Scenario B) of an object sliding on a track. For both recordings, the time interval between two successive positions is constant: $\tau = 50$ ms.



1. Determine the nature of the motion (uniform, accelerated, or decelerated) for **Scenario A**. Justify your answer based on the recording.
2. Determine the nature of the motion for **Scenario B**. Justify your answer.
3. **For Scenario A:** Calculate the average speed of the object between A_1 and A_5 in m/s.
4. **For Scenario B:** Calculate the instantaneous speed of the object at point B_3 in m/s.